Beyond Automated Feedback: *Advancing Educational Feedback through Human–AI Collaboration*

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The Workshop on Multi-modality and Multi-lingual Language Learning, in Conjunction with AACL-IJCNLP 2026

Generative Artificial Intelligence (GenAI)

Image: http://bit.ly/4o2yCR5

Ultimate Goal

Supporting human learning while benefiting from the power of AI

Key question & Today's talk

How can we empower educators to enhance their feedback practices in the age of AI?

Key question & Today's talk

How can we empower educators to enhance their feedback practices in the age of AI?

How can human–AI collaboration empower educators to enhance their feedback practices?

The very first step …

How are humans and GenAI working together in education?

The very first step …

How are humans and GenAI working together in education?

Modes of human–AI interaction in GenAI-supported education.

Theoretical Framework

RQ: What *modes of human–AI interaction* can be adopted to support feedback practices?

Liang, Z., Yang, K., Sha, L., Gašević, D., Yan, L., & Chen, G. (2026). A systematic review of generative AI in education: Empirical insights from a human–AI interaction perspective. *British Journal of Educational Technology*.

Cukurova, M. (2025). The interplay of learning, analytics and artificial intelligence in education: A vision for hybrid intelligence. British Journal of Educational Technology, 56(2), 469-488.

Theoretical Framework

Two dimensions shape human–AI interaction

- **Human control**
  - Who defines the goals?
  - Who interprets the information?
  - Who makes the final decision?
- **AI automation**
  - How independently does AI perform the task?
  - Does AI advise, generate or act?
  - How much human intervention is required?

Cukurova, M. (2025). The interplay of learning, analytics and artificial intelligence in education: A vision for hybrid intelligence. British Journal of Educational Technology, 56(2), 469-488.

Theoretical Framework

Human–AI interaction depends on control and automation

Internalization: AI informs; humans interpret and decide.

Hybrid intelligence: humans and AI jointly perform the task.

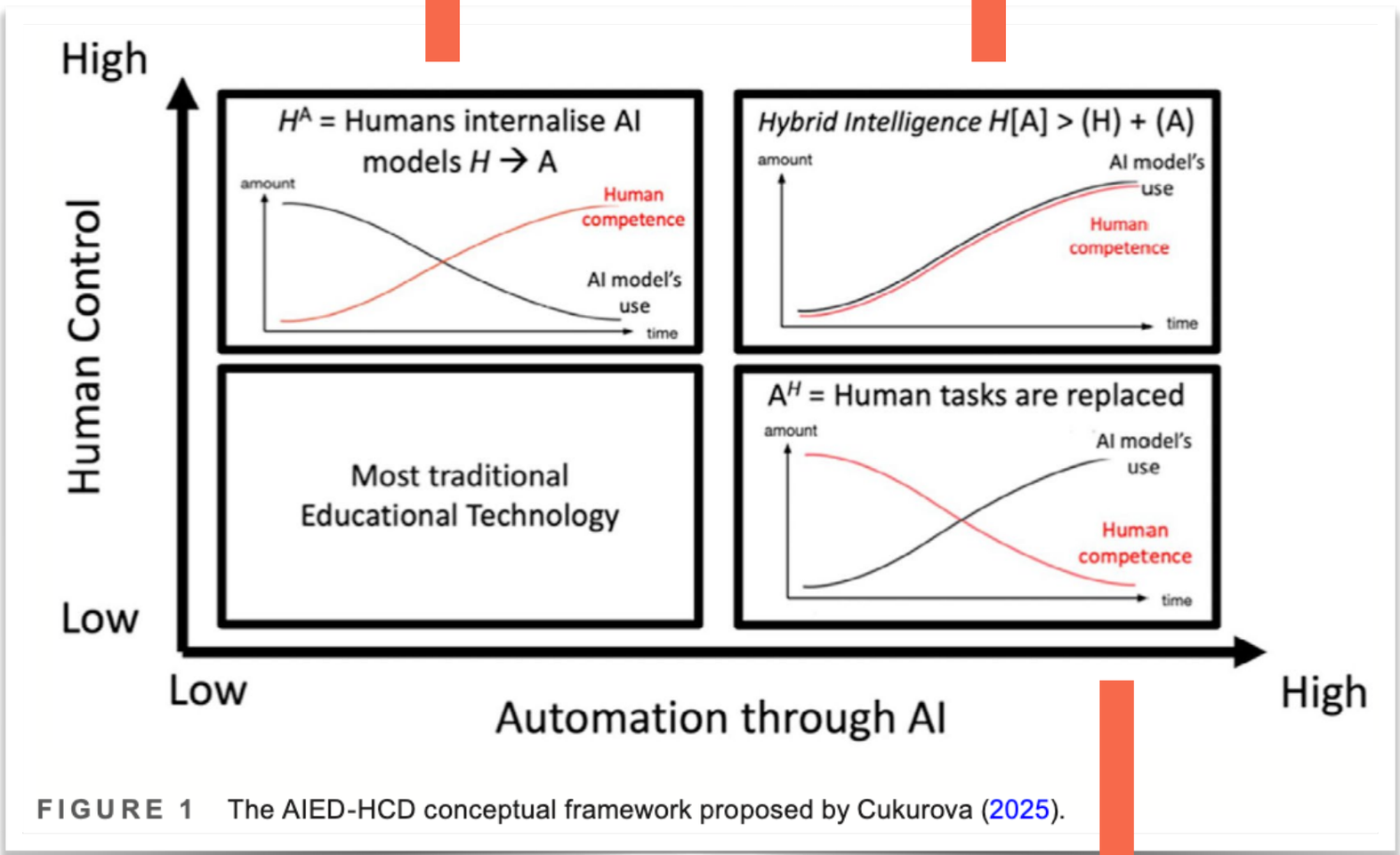

FIGURE 1 The AIED-HCD conceptual framework proposed by Cukurova (2025).

Externalization: AI performs tasks on humans' behalf.

Theoretical Framework

Implications for feedback practices

Interaction mode	Existing focus	Language-learning example
Externalization	AI generates feedback	AI automatically corrects grammar, vocabulary, and pronunciation.
Internalization	AI supports educator reflection	AI identifies missing components in teachers' feedback on students' English writing.
Hybrid intelligence	Humans and AI collaborate	Teachers guide AI in delivering personalized feedback based on learners' proficiency and learning needs.

Theoretical Framework

Implications for feedback practices

Interaction mode	Role of AI	Application to feedback
Externalization	Performs the task	AI generates feedback
Internalization	Supports human thinking	AI helps educators analyse and improve feedback
Hybrid intelligence	Collaborates with humans	Educators and AI co-create personalised feedback

To what extent can these modes support real-world educators' feedback practices?

Study 1: Automated Feedback Provision (Externalization)

RQ: To what extent can GenAI be used to provide quality feedback?

Dai, W., Lin, J., Jin, H., Li, T., Tsai, Y. S., Gašević, D., & Chen, G. (2023). Can large language models provide feedback to students? A case study on ChatGPT. In *2023 IEEE international conference on advanced learning technologies (ICALT)* (pp. 323-325). IEEE.

Dai, W., Tsai, Y. S., Lin, J., Aldino, A., Jin, H., Li, T., ... & Chen, G. (2024). Assessing the proficiency of large language models in automatic feedback generation: An evaluation study. *Computers and Education: Artificial Intelligence*, *7*, 100299.

Study 1: Automated Feedback Provision (Externalization)

Context & Dataset

Prompt Engineering

Evaluation

Study 1: Automated Feedback Provision (Externalization)

Context & Dataset

- 103 students' **data science** project proposals and the associated instructor-generated feedback.
- Five aspects in the marking rubric:
  1) Clear description of the goals of the project (**Goal**);
  2) Appropriateness of the topic to data science (**Topic**);
  3) Clear description of the business benefits (**Benefit**);
  4) Novelty/creativity (**Novelty**);
  5) Overall clarity of the report (**Clarity**).

Prompt Engineering

Evaluation

Study 1: Automated Feedback Provision (Externalization)

Context & Dataset

Prompt Engineering

Evaluation

- Both GPT-3.5 and GPT-4 were employed.
- Designed prompt:

"Please give feedback on the following text in terms of a clear description of the goals of the project, appropriateness of the topic to data science, a clear description of the business benefits, novelty/creativity and overall clarity of the report.

<INSERT THE TEXT OF A REPORT>"

Study 1: Automated Feedback Provision (Externalization)

Dataset

Prompt Engineering

Evaluation

- Readability
- Feedback Elements

Study 1: Automated Feedback Provision (Externalization)

Dataset

Prompt Engineering

Evaluation

- Readability
- Feedback elements

Three dimensions

- *Feeding up*: where students are going regarding their attainment of the learning goal;
- *Feeding back*: how students are going regarding assessments on student performance on a specific part of the task;
- *Feeding forward*: where students are going next regarding future improvements.

Four-level focuses

- *Task*: containing corrective information that indicates how well a task is performed;
- *Process*: suggesting strategies for completing the task;
- *Self-regulation*: addressing how students monitor their learning;
- *Self*: containing personal evaluations.

Hattie, J., & Timperley, H. (2007). The power of feedback. Review of educational research, 77(1), 81-112.

Study 1: Automated Feedback Provision (Externalization) - Results

- Readability

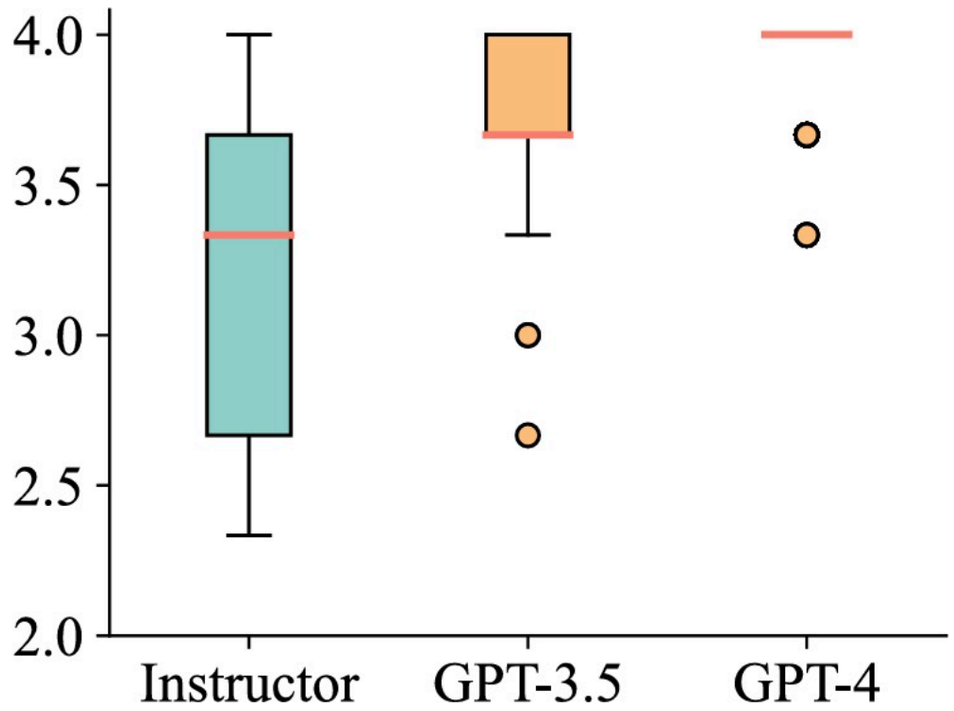

Fig. 1. Comparison of readability scores for feedback generated by human instructors, GPT-3.5 and GPT-4.

AI-generated feedback is more readable.

Study 1: Automated Feedback Provision (Externalization) - Results

- ## Feedback quality

Table 7
The comparison of the distribution of seven effective feedback components between the feedback provided by the human instructor, generated by GPT-3.5 and GPT-4. For each component, ANOVA tests were performed to measure the statistical difference between the three feedback generators. Subsequent post hoc comparisons were conducted using t-tests. Paris of feedback generators that demonstrated significant differences (p-value <= 0.05) are listed in the final column. For example, "GPT-4 vs {Instructor, GPT-3.5}" signifies that significant differences were identified between GPT-4 and Instructor, as well as between GPT-4 and GPT-3.5. The term "All" indicates that each pair of feedback generators exhibited a significant difference, whereas "None" denotes the absence of significant difference among all three feedback generators.

Components	Instructor		GPT-3.5		GPT-4		Significant results
	Quantity	Frequency	Quantity	Frequency	Quantity	Frequency	
Feeding up	6	5.83%	0	0%	97	94.17%	GPT-4 vs {Instructor, GPT-3.5}
Feeding back	101	98.06%	102	99.03%	103	100%	None
Feeding forward	93	90.29%	63	61.17%	100	97.09%	GPT-3.5 vs {Instructor, GPT-4}
Task	103	100%	103	100%	103	100%	None
Process	82	80%	57	55%	100	97.09%	All
Self-regulation	11	11%	0	0%	18	17.48%	GPT-3.5 vs {Instructor, GPT-4}
Self	25	24%	0	0%	0	0%	Instructor vs {GPT-3.5, GPT-4}

Study 1: Automated Feedback Provision (Externalization) - Results

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AI-generated feedback contains more quality feedback elements.

Study 1: Automated Feedback Provision (Externalization) - Results

- **Feedback quality**

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AI-generated feedback contains less feedback of personal evaluation.

Study 1: Automated Feedback Provision (Externalization) - Results

- ## Feedback quality

Table 8
Polarity of feedback generated by human instructors, GPT-3.5, and GPT-4 according to the five assessment aspects in the marking rubric. The precision, recall and F1 for measuring the agreement between GPT models and human assessors in terms of feedback polarity are included.

Aspects	Polarity	Quantity			Precision		Recall		F1	
		Instructor	GPT-3.5	GPT-4	GPT-3.5	GPT-4	GPT-3.5	GPT-4	GPT-3.5	GPT-4
Goal	**Positive**	5	95	74	0.04	0.05	0.80	0.80	0.08	0.10
	Negative	40	3	29	0.33	0.62	0.03	0.45	0.06	0.52
	None	58	5	0	0.40	0.00	0.03	0.00	0.06	0.00
Topic	**Positive**	86	95	99	0.84	0.85	0.93	0.98	0.88	0.91
	Negative	5	0	3	0.00	0.67	0.00	0.40	0.00	0.50
	None	12	8	1	0.13	0.00	0.08	0.00	0.10	0.00
Benefit	**Positive**	19	85	64	0.20	0.22	0.90	0.74	0.33	0.34
	Negative	59	5	37	0.60	0.73	0.05	0.46	0.09	0.56
	None	25	13	2	0.31	0.00	0.16	0.00	0.21	0.00
Novelty	**Positive**	31	52	64	0.35	0.44	0.58	0.90	0.44	0.59
	Negative	19	22	39	0.27	0.33	0.32	0.68	0.29	0.44
	None	53	29	0	0.59	0.00	0.32	0.00	0.41	0.00
Clarity	**Positive**	24	77	76	0.21	0.28	0.67	0.88	0.32	0.42
	Negative	11	17	27	0.29	0.19	0.46	0.46	0.36	0.27
	None	68	9	0	0.44	0.00	0.06	0.00	0.11	0.00

Study 1: Automated Feedback Provision (Externalization) - Results

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Topic	**Positive**	86	95	99	0.84	0.85	0.93	0.98	0.88	0.91
	Negative	5	0	3	0.00	0.67	0.00	0.40	0.00	0.50
	None	12	8	1	0.13	0.00	0.08	0.00	0.10	0.00
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	Negative	11	17	27	0.29	0.19	0.46	0.46	0.36	0.27
	None	68	9	0	0.44	0.00	0.06	0.00	0.11	0.00

AI often assesses student work incorrectly.

Study 1: Automated Feedback Provision (Externalization) - Results

- More recent studies

Studies	Limitations of AI-generated feedback
AlGhamdi et al. (2026). Leveraging prompt-based LLMs for automated scoring and feedback generation in higher education. *Computers & Education*, *243*, 105511.	**Affirmation** and **encouragement** were almost absent, and **learning-outcome alignment** was absent. Scoring errors were greater for lower-quality submissions.
Pecuchova, Benko, and Drlik (2026). Reimagining feedback through generative AI in engineering education. *Computers and Education: Artificial Intelligence*, *10*, 100574.	**AI** feedback showed **more repetition and less lexical diversity**; **teacher** comments offered **more specific reformulations, examples, and task-oriented guidance**.
Alsaiari et al. (2025). Emotionally enriched AI-generated feedback: Supporting student well-being without compromising learning. *Computers & Education*, *239*, 105363.	A **more supportive tone** in AI-generated feedback did **not** significantly **improve feedback engagement or revision quality.**

AI cannot always generate high-quality feedback.

So, instead of replacing educators, can it help them improve their own feedback?

AI cannot always generate high-quality feedback. So, instead of replacing educators, can it help them improve their own feedback?

From *Externalization* to *Internalization*

Study 2: AI-Supported Feedback Enhancement (Internalization)

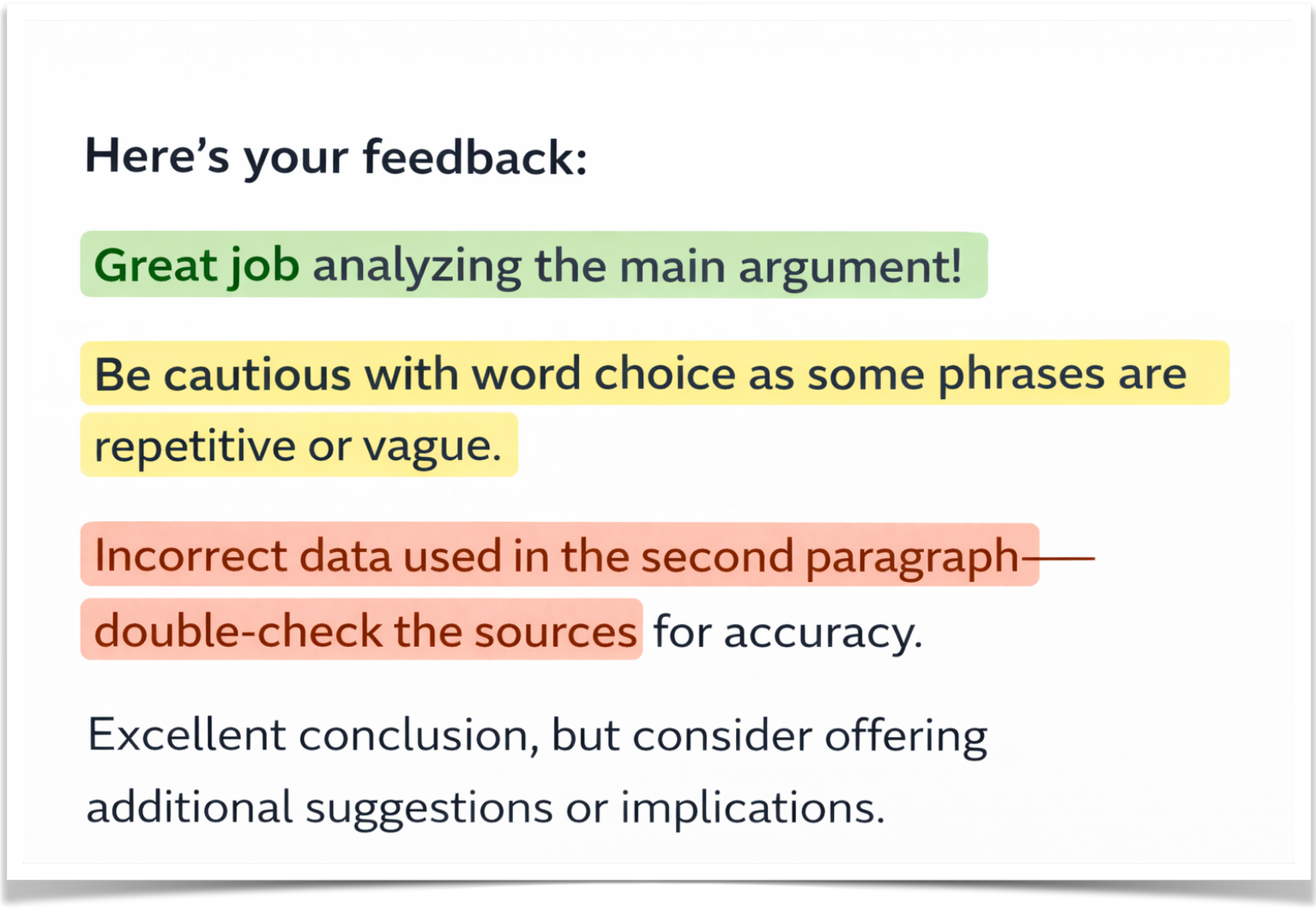
RQ: How can AI support educators in reflecting on and improving their own feedback?

Aldino, A. A., Tsai, Y. S., Mello, R. F., Gašević, D., & Chen, G. (2024). Enhancing Feedback Quality at Scale: Leveraging Machine Learning for Learner-Centered Feedback. *Computers and Education: Artificial Intelligence*, *7*, 100332.

Aldino, A. A., Maheshi, B., Li, Y., Zhou, Y., Tsai, Y. S., Gašević, D., & Chen, G. (2026). Enhancing learner-centered feedback with AI: teachers' practices and perceptions. Assessment & Evaluation in Higher Education, 1-18.

Study 2: AI-Supported Feedback Enhancement (Internalization)

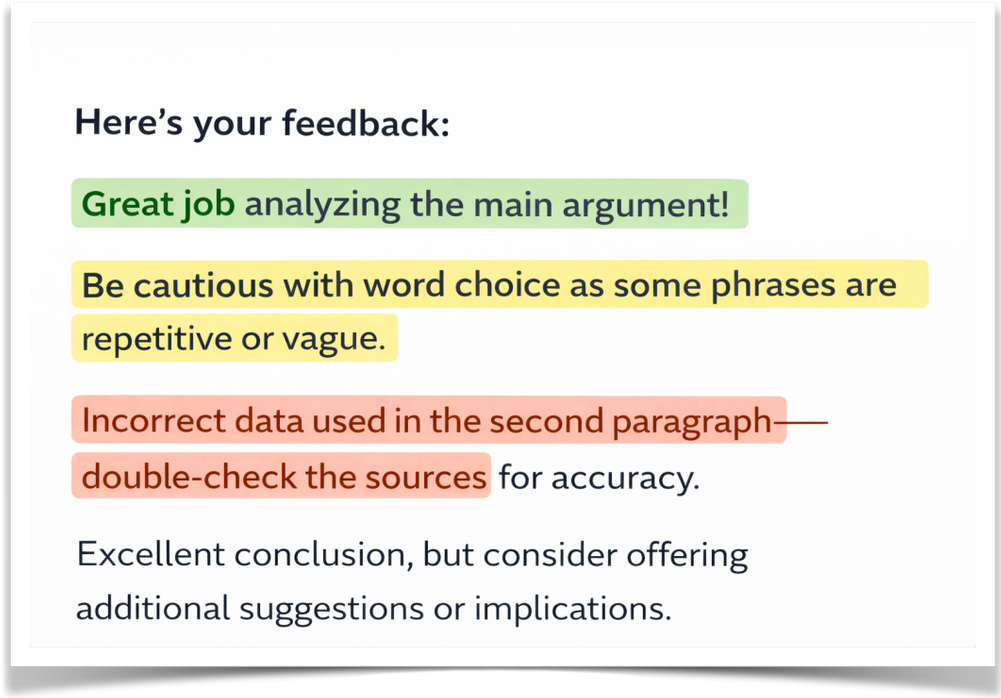
Ideally something like ….



Study 2: AI-Supported Feedback Enhancement (Internalization)

An educator-led workflow ...

Teacher drafts feedback
→ AI identifies components
→ AI highlights missing elements
→ Teacher accepts or rejects suggestions
→ ChatGPT enhances the draft
→ Teacher reviews and revises



Study 2: AI-Supported Feedback Enhancement (Internalization)

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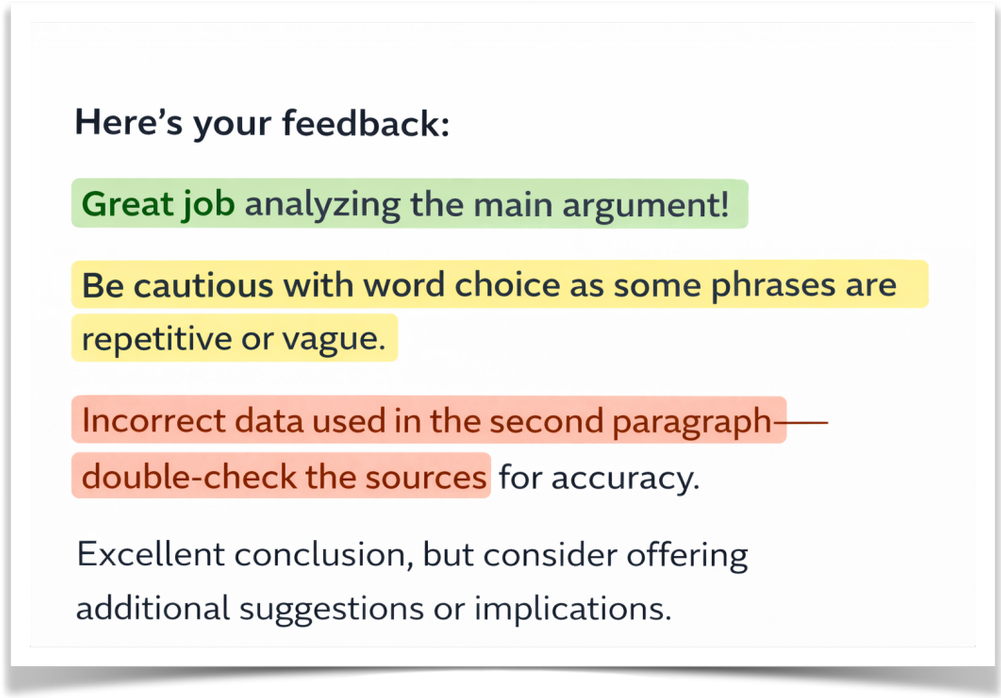

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

- 21 higher-education teachers with diverse disciplinary backgrounds and teaching experience
- Teachers watched a three-minute student presentation simulating a job interview.
- Each teacher independently drafted written feedback.

AI-Supported Feedback Analysis

Data Collection & Analysis

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

- AI identified learner-centred components in teacher feedback.

Table 1
Dimensions, Components (Labels), and Definitions of the Learner-Centered Feedback Framework.

Dimension	Component (Label)	Definition
Future Impact	FI_Future_Improvement	feedback which provides students with relevant, practical guidance to refine key elements of their future work, focusing on areas that show promise for development. (e.g. "Include at least one citation for each question and make sure you both cite in-line and correctly")
Future Impact	FI_Learning_Outcomes	feedback which provides students with targeted insights that facilitate the achievement of specific educational outcomes related to a subject matter. (e.g. "When you write your final teamwork reflection in assignment 2B, your analysis will need to more explicitly use the teamwork concepts")
Future Impact	FI_Skill_Development	feedback which fostering students' academic skills and strategies, contributing to their educational and professional advancement. (e.g. "Also please make sure to spell check your work before submission")
Sensemaking	S_Performance_Summary	feedback which provides an all-encompassing assessment of a student's performance, aligning their strengths and weaknesses with the established educational goals and assessment criteria. (e.g. "In general, this report lacks the level of definition and description required from a masters student")
Sensemaking	S_Strengths_Weaknesses	feedback which provides commendable aspects and improvement areas in a student's work, covering key facets such as content accuracy, grammatical precision, and structural integrity. (e.g. "Referencing is not to APA standards")
Agency	A_Active_Role	feedback which encourages students to take an engaged and proactive role in their learning journey, promoting dialogue with educators and exploration of additional educational resources. (e.g. "I would recommend googling "Gibbs cycle of reflection" and try to apply that to your diaries")
Agency	A_Affirm_Encourage	feedback which affirm the student's successful efforts in their assignments and encourages continued growth and improvement in their future tasks. (e.g. "Hi, Great job on submitting your first reflective diary for the semester")
Agency	A_Strengthen_Relationship	feedback which seeks to enhance the educational bond between the teacher and the student through nurturing and supportive interactions. (e.g. "I really enjoyed teaching you")

Ryan, T., Henderson, M., Ryan, K., & Kennedy, G. (2023). Identifying the components of effective learner-centred feedback information. Teaching in Higher Education, 28, 1565–1582.

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

- AI identified learner-centred components in teacher feedback.
- About 9,990 feedback sentences from 95 courses at Monash University were labelled according to the learner-centred feedback framework.
- A BERT-based classifier identified which components were present—and which might be missing.

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

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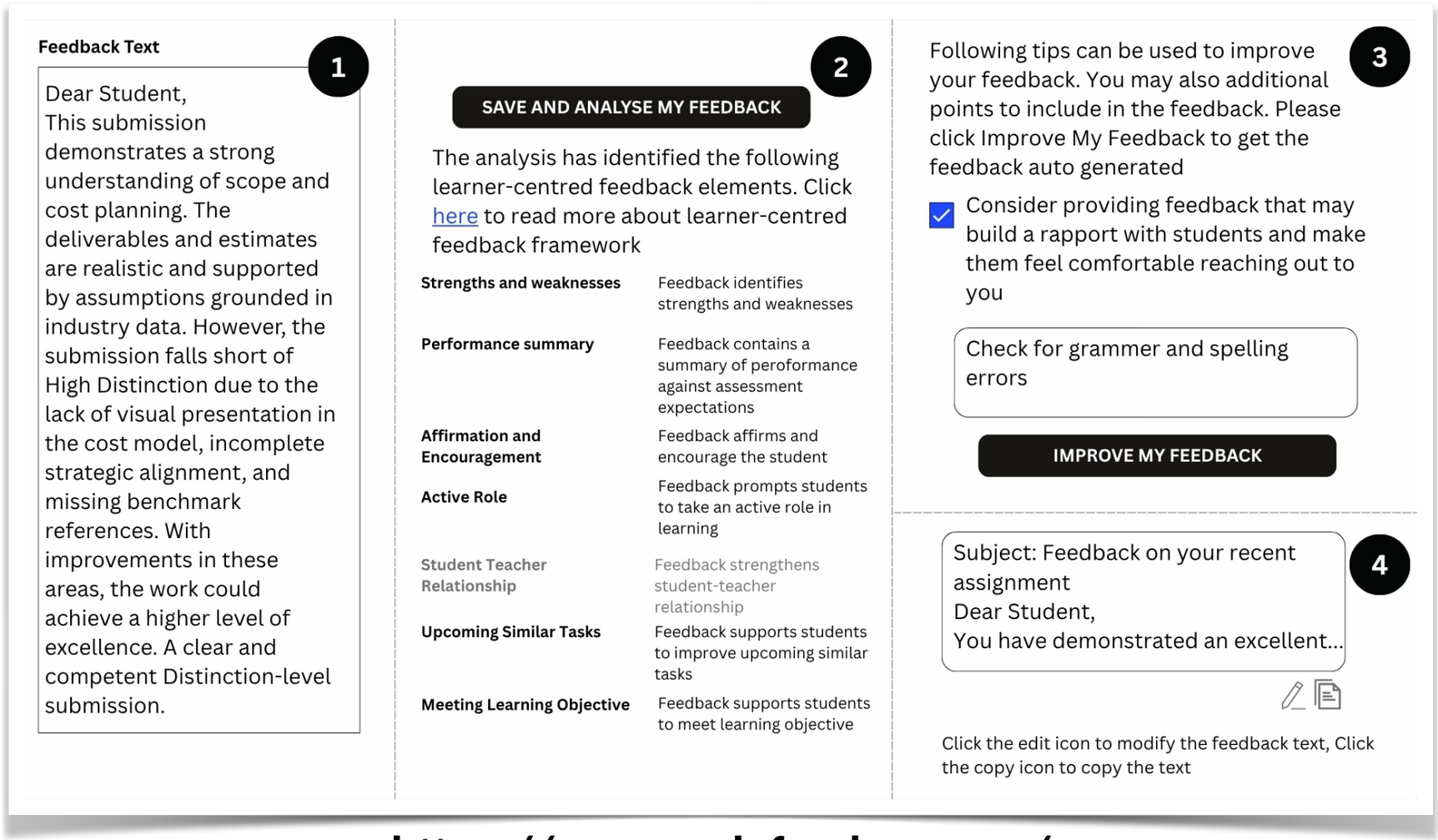
- About 9,990 feedback sentences from 95 courses at Monash University were labelled according to the learner-centred feedback framework.
- A BERT-based classifier identified which components were present—and which might be missing.
- The classifier was incorporated into **PolyFeed**, a feedback system to support educators to provide and manage quality feedback using AI-powered learning analytics.

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis



https://www.polyfeed.com.au/

Bhagya Maheshi, Hiruni Palihena, Flora Yi-Joon Jin, Ahmad Aldino, Thomas Ng, Ziyi Wang, Sachini Samaraweera, Danijela Gasevic, Nicola Charwat, Philip Wing Keung Chan, Guanliang Chen, Sadia Nawaz, Roberto Martinez-Maldonado, Dragan Gaševic, Yi-Shan Tsai. PolyFeed for Educators: A Feedback Analytics Tool to Facilitate Effective Feedback Processes. In Proceedings of the 16th International Conference on Learning Analytics and Knowledge, 2026. (LAK 2026)

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

Four forms of data:

- Original teacher-written feedback
- AI classifications and teacher acceptance or rejection decisions
- ChatGPT-enhanced and teacher-revised feedback
- Semi-structured interviews with all 21 teachers

Study 2: AI-Supported Feedback Enhancement (Internalization)

Study Context

AI-Supported Feedback Analysis

Data Collection & Analysis

Four forms of data:

- Original teacher-written feedback
- AI classifications and teacher acceptance or rejection decisions
- ChatGPT-enhanced and teacher-revised feedback
- Semi-structured interviews with all 21 teachers

Three layers of teacher–AI interaction were examined

1. Responding to AI classifications
   - Which missing-component suggestions did teachers accept or reject?
2. Revision actions
   - What did teachers add, edit or remove from the AI-enhanced draft?
3. Nature of the revisions
   - Did teachers adjust content, tone and brevity?

Study 2: AI-Supported Feedback Enhancement (Internalization) - Results

Teachers frequently acted on AI prompts about missing components

100%	Affirmation & encouragement	6 of 6 accepted
86%	Student–teacher relationship	12 of 14 accepted
80%	Meeting learning objectives	16 of 20 accepted

Study 2: AI-Supported Feedback Enhancement (Internalization) - Results

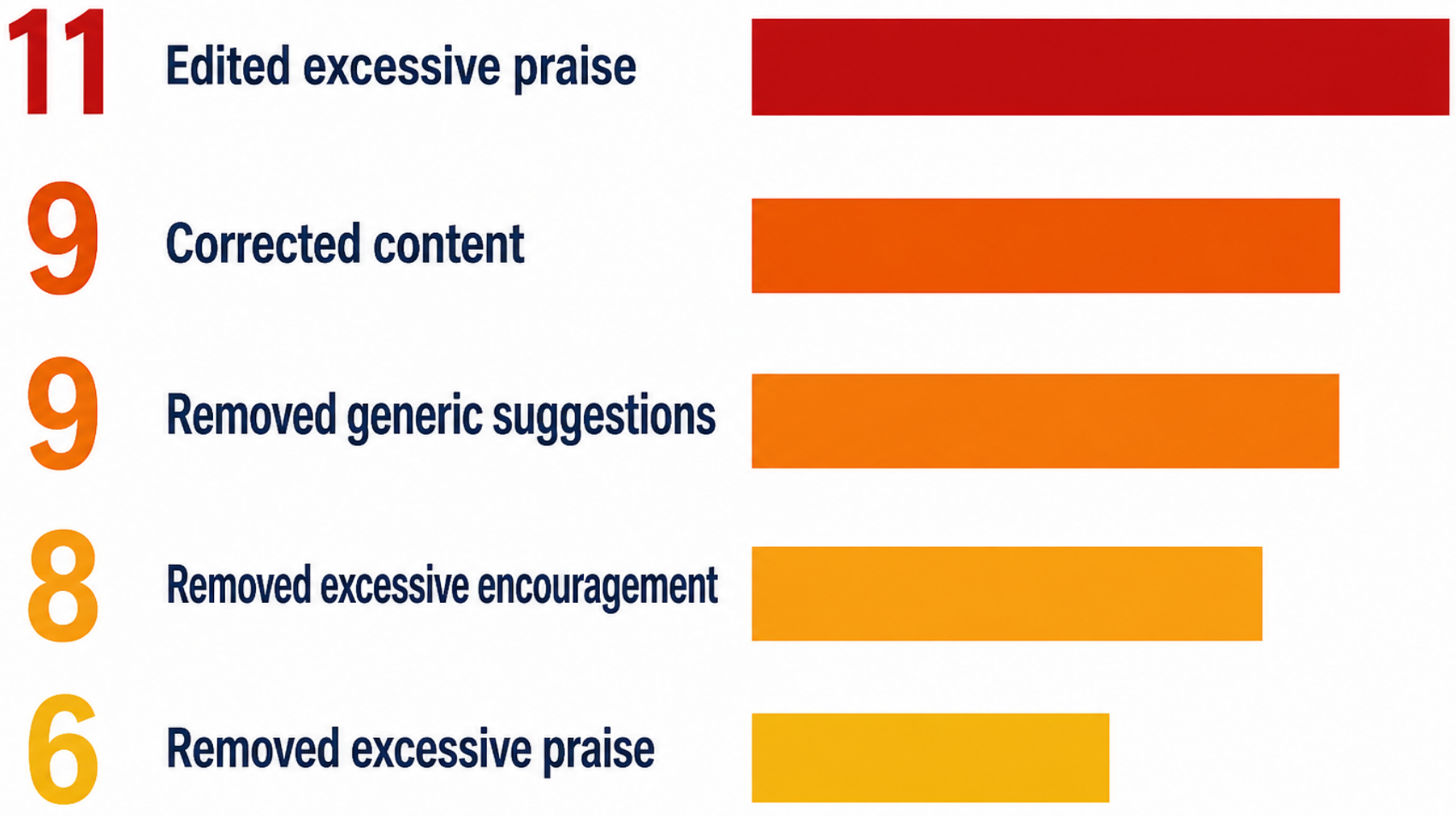
Teachers did not passively accept AI-enhanced feedback

12 of 21 teachers
revised the AI-enhanced feedback

32	27	8
Edits	Removals	Additions

Study 2: AI-Supported Feedback Enhancement (Internalization) - Results

Teachers used professional judgement to calibrate AI output



Study 2: AI-Supported Feedback Enhancement (Internalization) - Results

Interviews examined perceived benefits and challenges

Perceived benefits

14	Promoted reflection
11	Improved language & structure
10	Identified missing components
2	Saved time

Perceived challenges

9	Required human editing
7	Inconsistent tone
5	Potential misinformation
5	Trust concerns

Can we preserve certain level of human control while increasing the level of AI automation?

Can we preserve certain level of human control while increasing the level of AI automation?

Hybrid Intelligence

Study 3: Edvance (Hybrid Intelligence)

RQ: How can GenAI be used to provide actionable, personalized, and adaptive feedback to students throughout the semester?

Liang, Z., Sha, L., Tsai, Y. S., Gašević, D., & Chen, G. (2024, July). Towards the automated generation of readily applicable personalised feedback in education. In *International Conference on Artificial Intelligence in Education* (pp. 75-88). Cham: Springer Nature Switzerland.

Li, Y., Liang, Z., Gašević, D., & Chen, G.. Edvance: An AI-Powered Tool for Delivering Actionable, Personalised, and Adaptive Feedback at Scale. In I*nternational Conference on Learning Analytics and Knowledge*, 2026.

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

Study 3: Edvance (Hybrid Intelligence)

Course Context

- IT course at Monash University; log trace data from Moodle
- Courses were carried out in a weekly manner (12 weeks per offering).

Feedback Generation

Evaluation

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

- Step 1: *Predictive* AI to identify at-risk students on a *weekly* basis.
  - Feature engineering

Table 1. A summary of the feature used in this study. * represents that the feature value is a binary value

Category	Feature Description
Course Contents	The fraction of specific types of course materials (Including: **F1**:Pre-workshop, **F2**:Workshop, **F3**:Applied, **F4**:Post-workshop) from week$_n$ accessed by the student
	The total amount of time the student spent viewing specific types of course materials (Including: **F5**:Pre-workshop, **F6**:Workshop, **F7**:Applied, **F8**:Post-workshop) from week$_n$
	F9 : The total minutes a student has dedicated to viewing slides during week$_n$
	F10: The number of slides a student has viewed during week$_n$
Forum	**F11**: The total minutes a student has dedicated to viewing the thread posted on the discussion forum during week$_n$
	The ratio of specific forum-related events (Including: **F12**:View thread, **F13**:Post thread, **F14**:Post comment, **F15**:Post answer) to the total forum-related events during week$_n$
	The number of specific forum-related events (Including: **F16**:View thread, **F17**:Post thread, **F18**:Post comment, **F19**:Post answer) presented by the student during week$_n$
Assessment	Indicate whether the student attempted the assessment (Assessment types, including: **F20**:Quiz, **F21**:Challenge) during week$_n$*. '1' to represent attempted and '0' otherwise
	The number of times a student has attempted the assessment (Assessment types, including: **F22**:Quiz, **F23**:Challenge) during week$_n$
	The score of the assessment (Assessment types, including: **F24**:Quiz, **F25**:Challenge, **F26**:Assignment 1) attempted by a student during week$_n$. (Assignment score will only be available at the end of Week 7; therefore, it will be set as 0 from Week 1 to Week 6)
	The average time interval between two attempted assessment (Assessment types, including: **F27**:Quiz, **F28**:Challenge) during Week$_n$ in minutes

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

- Step 1: *Predictive* AI to identify at-risk students on a *weekly* basis.
  - Feature engineering

Table 1. A summary of the feature used in this study. * represents that the feature value is a binary value

Category	Feature Description
Course Contents	The fraction of specific types of course materials (Including: **F1**:Pre-workshop, **F2**:Workshop, **F3**:Applied, **F4**:Post-workshop) from week$_n$ accessed by the student
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	The score of the assessment (Asse… attempted by a student during we… it will be set as 0 from Week 1 to …
	The average time interval between … …:Quiz, **F28**:Challenge) during Week$_n$ in m…

This is where we ensure Human Control

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

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- Step 1: *Predictive* AI to identify at-risk students on a *weekly* basis.
  - Feature engineering
  - Predictive models: Naive Bayes, SVM, Random Forest, XGBoost, and Multi-layer perceptron

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

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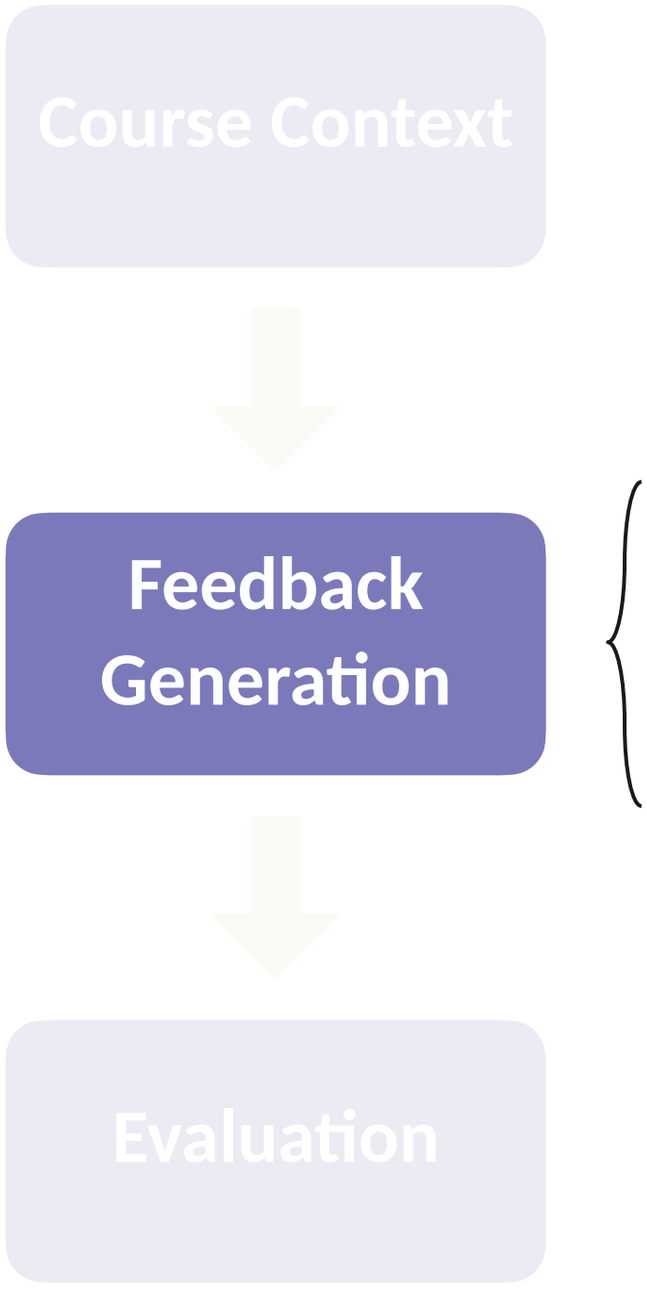
- Step 1: *Predictive* AI to identify at-risk students on a *weekly* basis.
  - Feature engineering
  - Predictive models: Naive Bayes, SVM, Random Forest, XGBoost, and Multi-layer perceptron
  - What about **scalability**?

Study 3: Edvance (Hybrid Intelligence)



- Step 1: *Predictive* AI to identify at-risk students on a *weekly* basis.
  - Feature engineering
  - Predictive models: Naive Bayes, SVM, Random Forest, XGBoost, and Multi-layer perceptron
  - What about **scalability**? **AutoML** can help!

Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

- Step 2: *Prescriptive* AI to provide actionable feedback
  - Diverse Counterfactual Explanations (DiCE)

Study 3: Edvance (Hybrid Intelligence)

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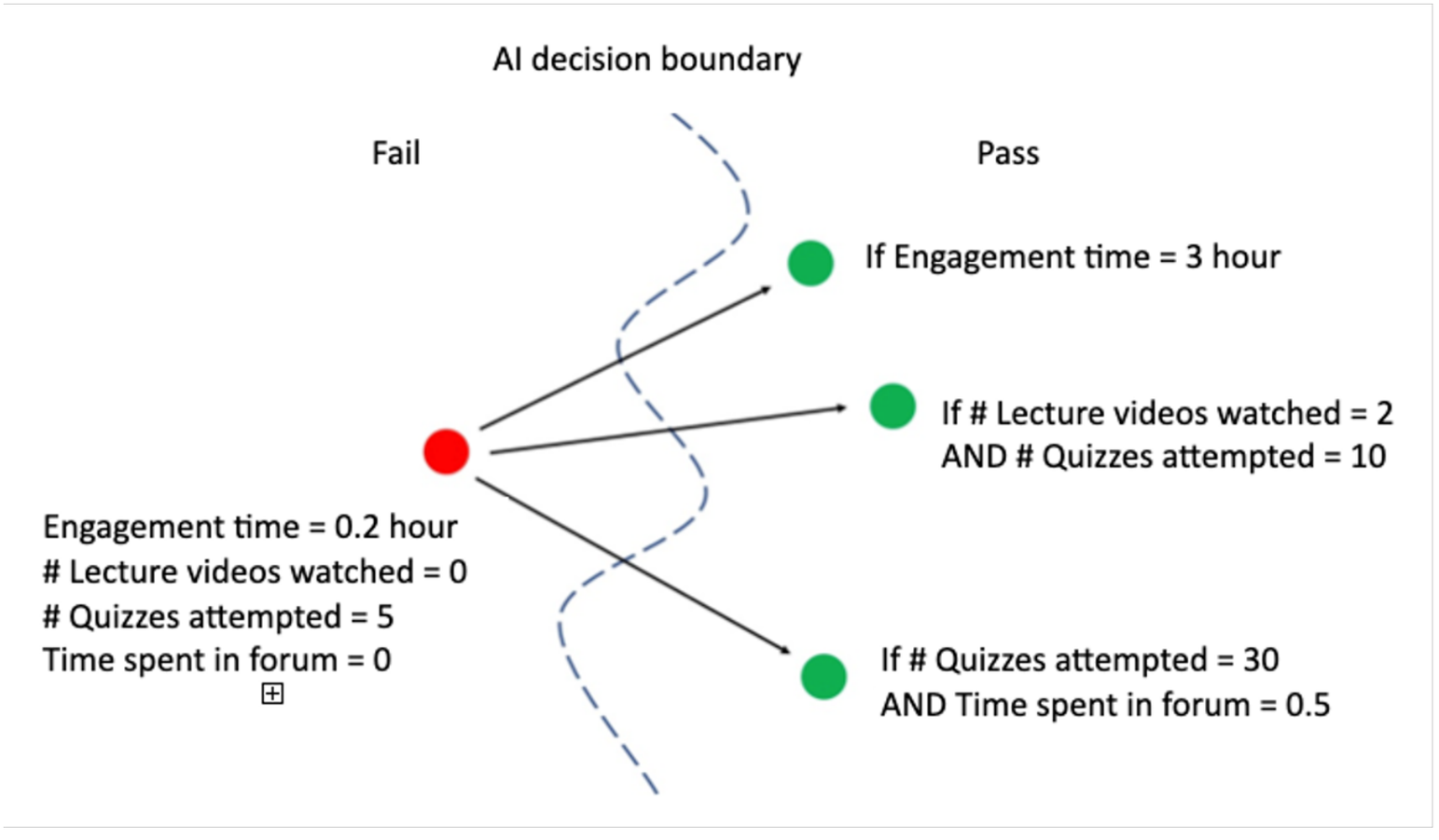

Study 3: Edvance (Hybrid Intelligence)

Course Context

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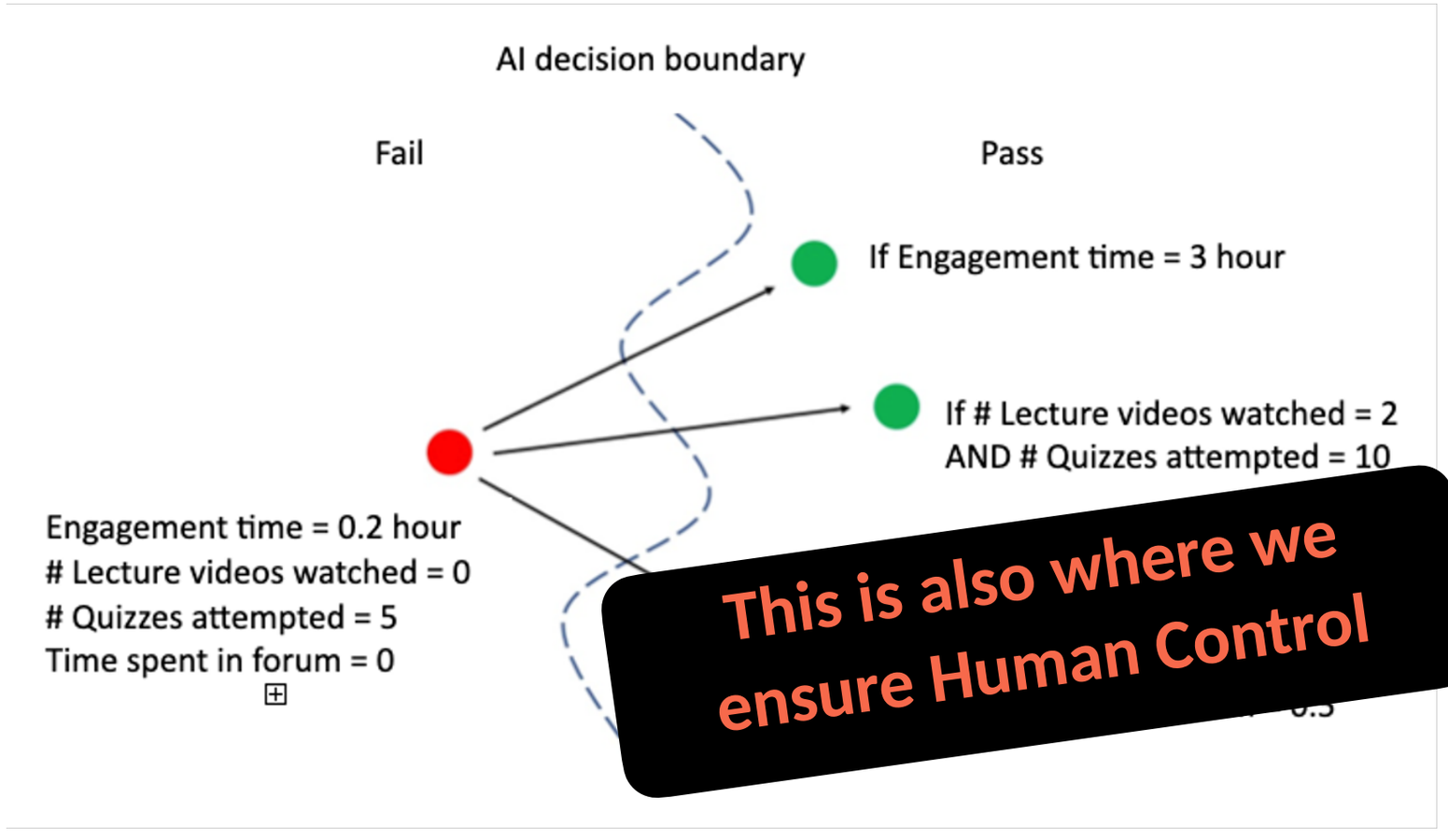


Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

- Step 3: *Generative AI* (GPT-5) to provide actionable feedback, guided by the learner-centred feedback framework

Study 3: Edvance (Hybrid Intelligence)

- Step 3: *Generative AI* (GPT-5) to provide actionable feedback, guided by the learner-centred feedback framework

"Please act as a teacher to provide feedback for students. The feedback must adhere to the following rules: [Learner-centered feedback rules]. The feedback is based on the students' higher-than-median learning activities: [Commendable learning activities], and the under-performed learning activities with the corresponding suggestions: [Counterfactual feature value changes]."

Study 3: Edvance (Hybrid Intelligence)

Course Context

↓

Feedback Generation

↓

Evaluation

- Step 3: *Generative AI* (GPT-5) to provide actionable feedback, guided by the learner-centred feedback framework
  - When generating advice, try to relate it to the corresponding *weekly topic* or *learning outcome*.
  - Each advice should not exceed *50 words*.
  - Maintain a *supportive*, *professional*, and *encouraging* tone.

Study 3: Edvance (Hybrid Intelligence)

- Step 3: *Generative AI* (GPT-5) to provide actionable feedback, guided by the learner-centred feedback framework

Feedback examples:

- *"Your current engagement in Week 2 Workshop activities is 0 hours. Aim to increase this to 0.5 hours to effectively grasp the concepts of logical statements and conditional execution."*
- *"You answered 2 quiz questions correctly in Week 2, but should target 4 to strengthen your grasp of conditional execution and logical expressions."*

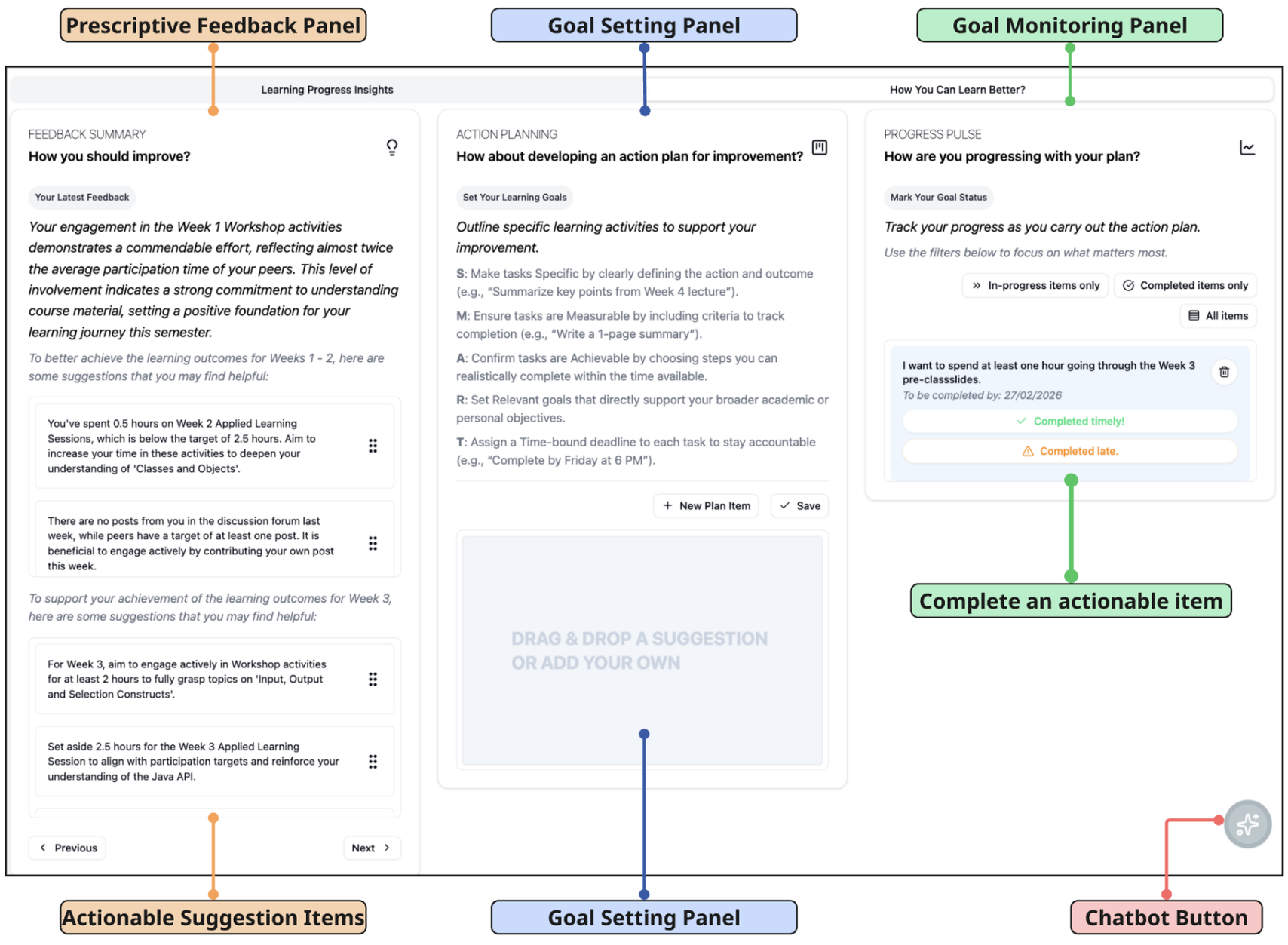
More technical details: https://docs.google.com/presentation/d/1VrpHz_NaXtCMcy2VSlz40DScGIf4laXtfv56CIFCEhI/edit?usp=sharing

Study 3: Edvance (Hybrid Intelligence)

Edvance

- *Developed as a plugin, easily integrated with Moodle, Ed, etc.*

Student-facing interface



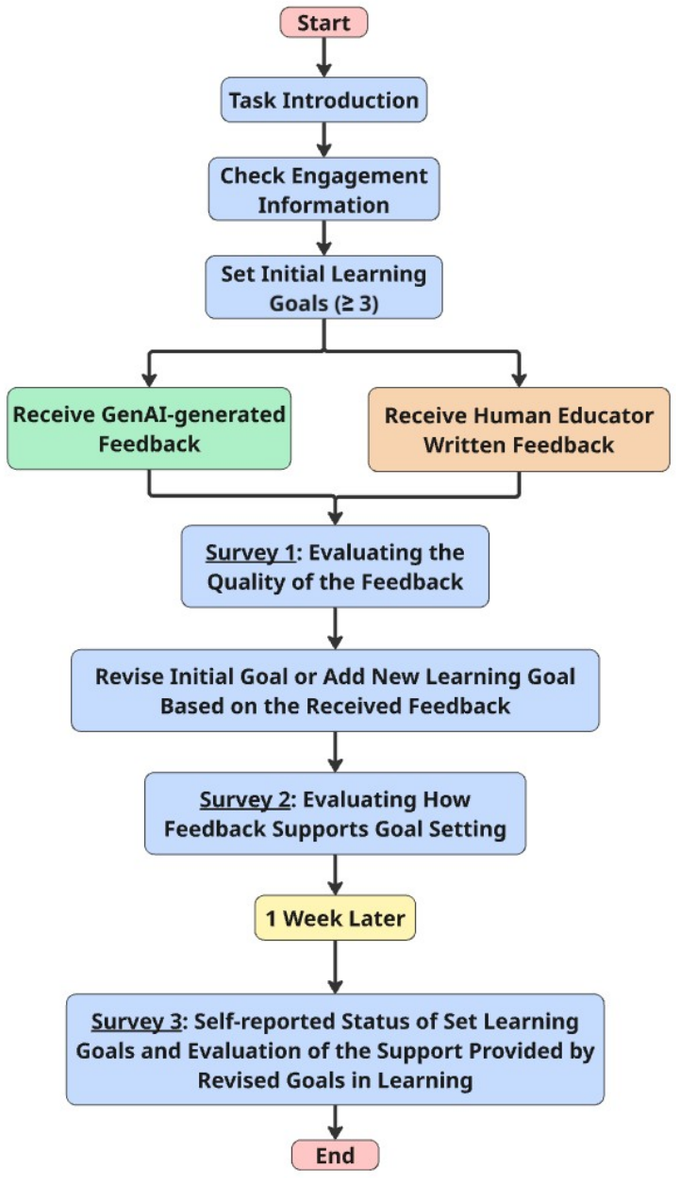
Study 3: Edvance (Hybrid Intelligence)

Course Context

Feedback Generation

Evaluation

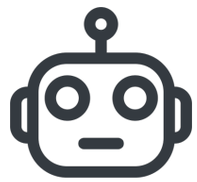
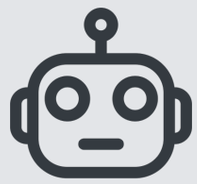
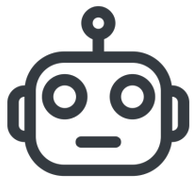
- A randomised controlled lab study with 13 participants



Study 3: Edvance (Hybrid Intelligence)

Edvance

- Preliminary findings include …

Feedback Quality		≈ generally comparable, except for impact on **motivation**	
Goal Setting		< Specific, time-bounded	
Goal Completion		≈ generally comparable	

FINAL REMARKS

- **AI automation can scale feedback provision,** but quality, accuracy, tone and contextual relevance still require human oversight.
- **Internalization can strengthen educator practice:** AI makes patterns and gaps visible, while educators retain professional judgement and authentic voice.
- **Hybrid intelligence offers the way forward:** combine AI's scalability and personalisation with educators' pedagogical and relational expertise.

MULTIMODAL AND MULTILINGUAL LANGUAGE LEARNING

- **Multimodal learner modelling**: How can we integrate writing, speech, and interaction data to identify learners' evolving needs?
- **Multilingual feedback adaptation**: How can feedback be tailored to different linguistic backgrounds, proficiency levels, and target languages, including low-resource languages?
- **Human–AI feedback orchestration**: How can teachers and AI collaborate to determine when, what, and how feedback should be delivered?

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- **Human–AI feedback orchestration**: How can teachers and AI collaborate to determine when, what, and how feedback should be delivered?
- **Success should be measured by human learning**, not simply by the amount of work automated.

Thanks! Any questions?